



CHATBOT BASED ON DATA SCIENCE ENQUIRY

Project Documentation

Submitted By:

Chetana Bhansali

Dristi Singh

Vidit Kumar Singh

Sonakshi Priyadarshini

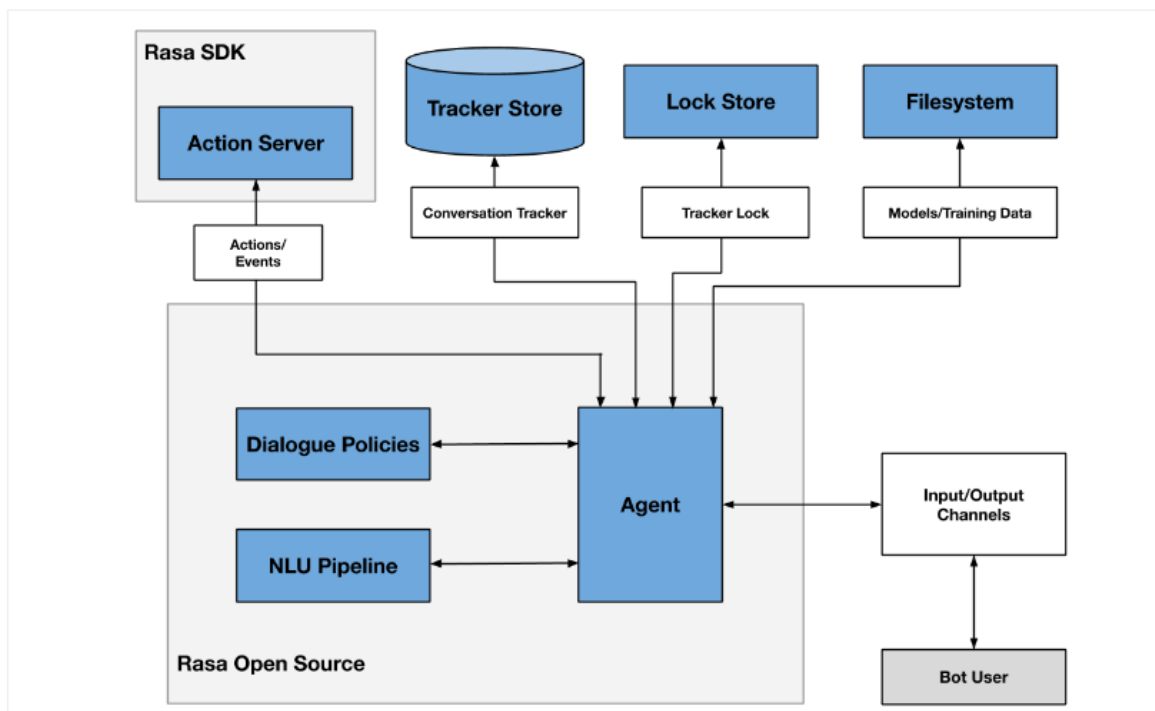
Chatbot Based On Data Science Enquiry

The “**Chatbot for Data Science Inquiries**” project is designed to create an intelligent conversational system that leverages advanced data science methodologies to assist users with queries related to data science concepts, tools, and techniques. This chatbot will harness the power of Natural Language Processing (NLP) to interpret user input, extract meaningful insights, and deliver precise answers by accessing a structured knowledge base or database. By incorporating machine learning algorithms for text classification, information retrieval, and sentiment analysis, the system aims to provide responses that are both personalized and contextually relevant. Ultimately, the project seeks to enhance the learning experience and streamline access to information in the dynamic and complex domain of data science.

Objective

The objective of this project is to develop an intelligent chatbot capable of handling data science-related inquiries by leveraging the RASA NLU and RASA Core frameworks. Through this project, users will gain practical knowledge of natural language understanding (NLU), text data processing, and conversational AI design. The project aims to provide hands-on experience in training text-based models, implementing intent classification and entity extraction, and building context-aware dialogue systems using the RASA ecosystem.

Technical Architecture



Project Flow

- The user interacts with the RASA Model through Command Prompt with a Text Message.
- Rasa NLU will learn from the labelled examples and optimize its model parameters.
- It understands user queries and extracting intents and entities, and generates the Responses.

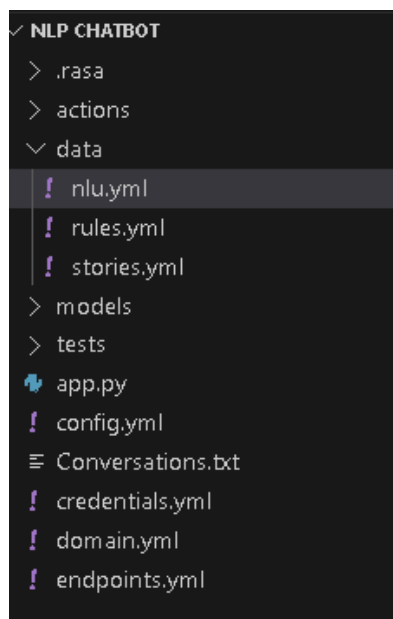
To accomplish this, we must complete all the activities and tasks listed below.

- Data Collection - Gather labelled data that consists of user queries, intents (the purpose or goal of the query), and entities (specific information mentioned in the query).
- Data Preprocessing - Structure the collected data in the appropriate training data format, such as Markdown or YAML. Each training example should include the user query, intent label, and entity annotations.
- Model Building - Use the prepared training data and the defined pipeline configuration to train the NLU model. Rasa NLU will learn from the labelled examples and optimize its model parameters.
- Testing and Deployment - Test the integrated chatbot by interacting with it using different user queries and scenarios. Once you are satisfied with the results, deploy the chatbot to your preferred deployment environment, such as a web server or a messaging platform.

Prior Knowledge

- Basic understanding of Python programming
- Fundamentals of machine learning algorithms
- Core concepts of Natural Language Processing (NLP)
- Experience with text data preprocessing (e.g., tokenization, stopwords removal)
- Familiarity with training and evaluating models
- Understanding of intent classification and entity recognition
- Awareness of RASA framework components (NLU & Core)
- Basic knowledge of YAML configuration files
- Understanding of chatbot workflows and dialogue management
- Familiarity with command-line interfaces and virtual environments
- Basic version control using Git

Project Structure



- config.yml: This file contains the configuration for the NLU pipeline. It defines the sequence of processing components used by Rasa NLU.
- nlu.yml: This file contains the training data for the NLU model. It consists of labeled examples of user queries, intents, and entities.
- domain.yml: This file defines the domain of your chatbot. It includes information about intents, entities, actions, responses, and templates.
- models: This directory contains the trained NLU model files generated after running the training process. It includes files like nlu.tar.gz or nlu-<timestamp>.tar.gz, which encapsulate the trained model parameters and can be used for inference.
- tests: This directory is used for storing test files and evaluation data. It can include test scripts or evaluation datasets to assess the performance of the NLU model.

- **credentials.yml:** If your chatbot integrates with external services, such as APIs or databases, you can store the credentials for those services in this file. It ensures secure access to the required resources.
- **endpoints.yml:** This file is used to configure the endpoints for connecting your chatbot with other services or platforms. It can include settings for platforms like Slack or Facebook Messenger, as well as the URL of the chatbot server.
- **actions:** This directory is used to define custom actions that your chatbot can perform. Actions can include API calls, database operations, or any other backend logic. Custom action files are typically implemented as Python scripts.

Milestone 1: Define Problem / Problem Understanding

Activity 1.1: Specify the Business Problem

In the current digital learning landscape, individuals often seek quick and reliable information on data science concepts, tools, and techniques. However, navigating through vast resources can be time-consuming and inefficient. This project aims to develop a chatbot that can intelligently respond to user queries related to data science using Natural Language Processing (NLP) and machine learning. The chatbot will understand, interpret, and answer domain-specific questions, serving as a virtual assistant for students, professionals, and enthusiasts looking to enhance their knowledge in data science.

Activity 1.2: Business Requirements

A chatbot designed for data science inquiries should fulfil several business and technical needs to be effective and scalable. Key requirements include:

- **Accuracy and Relevance:** The chatbot must provide correct and up-to-date responses based on the latest developments and reliable sources in the data science domain.
- **Natural Language Understanding (NLU):** The system should accurately interpret user intents and extract relevant entities from diverse input formats.
- **Scalability:** It should be designed to handle an increasing number of user interactions and queries as the user base grows.
- **Extensibility:** The system should allow for easy addition of new intents, responses, and domain knowledge over time.
- **Compliance and Data Security:** All data processing should comply with privacy and data security regulations (e.g., GDPR, if deployed publicly).
- **User-Friendly Interface:** The chatbot should have an intuitive and accessible interface to encourage usage across a range of users with varying technical backgrounds.
- **Performance and Speed:** It should respond to queries in real-time with minimal latency.
- **Logging and Feedback Mechanism:** Maintain logs and allow user feedback to improve chatbot responses and training data.

Activity 1.3: Research Background

The development of chatbots for educational and domain-specific purposes has seen a significant rise in recent years, particularly due to advancements in Natural Language Processing (NLP) and machine learning. Several studies have explored the use of chatbots in areas such as healthcare, customer service, and e-learning, establishing their effectiveness in providing instant and reliable information to users.

In the context of data science education, literature reveals that users often face challenges in understanding complex concepts due to the lack of interactive and real-time guidance. To address this, chatbot systems have been explored as potential tools for delivering knowledge in an engaging and accessible manner. Chatbots powered by frameworks like RASA have proven effective in combining intent recognition, entity extraction, and dialogue management to simulate human-like conversations.

Studies such as “A Review on Chatbot Design and Implementation Techniques” (2020) emphasize the importance of choosing the right architecture—rule-based vs. AI-based—and underline the role of NLP in understanding user intent. Research on open-domain vs. closed-domain chatbots highlights that domain-specific bots, like the one proposed in this project, often achieve higher accuracy due to their focused scope.

In addition, literature on the RASA framework shows that its modular design, with components like RASA NLU and RASA Core, allows developers to build scalable and context-aware chatbots. RASA’s open-source nature also makes it a popular choice for academic and industry projects focused on conversational AI.

Moreover, academic works suggest that incorporating user feedback, training with real-world queries, and iterative model improvements are key to building an effective educational chatbot. Techniques such as supervised learning, sentiment analysis, and contextual embeddings (e.g., BERT) are also commonly discussed as means to enhance chatbot intelligence and relevance.

Overall, existing literature supports the feasibility and value of developing a chatbot for data science queries. However, there remains a gap in tools specifically tailored to the data science learning journey—covering technical queries, tool usage, and concept explanations—which this project aims to bridge.

Activity 1.4: Social or Business Impact

- Social Impact – Empowered Learning and Accessibility:

The chatbot enhances accessibility to complex data science topics, enabling students and professionals to learn at their own pace without relying solely on lengthy documentation or external help. It promotes self-learning and boosts confidence in technical domains.

- Business Impact – Scalable Support and Training Tool:

The chatbot can be integrated into online learning platforms or corporate training environments to provide scalable, on-demand support. Organizations can reduce the burden on human instructors or support teams while ensuring consistent, high-quality responses to user inquiries.

Milestone 2: Environmental Setup

- In this Milestone we focus on preparing your development environment to work with Rasa NLU. It involves installing Python, creating a virtual environment, and installing the necessary dependencies.
- Create a Virtual Environment: Create a separate virtual environment for your Rasa NLU project to isolate dependencies. Open a terminal or command prompt and navigate to your project directory. Use a virtual environment tool 'conda' to create a new virtual environment and Run the Command.
- Activate the New Virtual Environment.

Activity 2.1: Install Rasa and Dependencies

Once the virtual environment is active, you can install Rasa and its dependencies using pip, the Python package manager.

```
(base) C:\Users\DELL>conda --version
conda 25.5.1

(base) C:\Users\DELL>conda create -n rasa_env python=3.8
Retrieving notices: done
Channels:
- defaults
Platform: win-64
Collecting package metadata (repodata.json): done
Solving environment: done

## Package Plan ##

  environment location: C:\Users\DELL\anaconda3\envs\rasa_env
  added / updated specs:
    - python=3.8

The following packages will be downloaded:



| package                    | build          | size           |
|----------------------------|----------------|----------------|
| ca-certificates-2025.2.25  | haa95532_0     | 130 KB         |
| libffi-3.4.4               | hd77b12b_1     | 122 KB         |
| openssl-3.0.16             | h37729d1_0     | 7.8 MB         |
| pip-24.2                   | py38haa95532_0 | 2.4 MB         |
| python-3.8.20              | h8285438_0     | 19.4 MB        |
| setuptools-75.1.0          | py38haa95532_0 | 1.6 MB         |
| sqlite-3.45.3              | h2bbff1b_0     | 975 KB         |
| vc-14.42                   | haa95532_5     | 11 KB          |
| vs2015_runtime-14.42.34433 | hbf6602d_5     | 1.2 MB         |
| wheel-0.44.0               | py38haa95532_0 | 137 KB         |
| <b>Total:</b>              |                | <b>33.7 MB</b> |



The following NEW packages will be INSTALLED:



| package         | pkgs/main/win-64                                       | size |
|-----------------|--------------------------------------------------------|------|
| ca-certificates | pkgs/main/win-64::ca-certificates-2025.2.25-haa95532_0 |      |
| libffi          | pkgs/main/win-64::libffi-3.4.4-hd77b12b_1              |      |
| openssl         | pkgs/main/win-64::openssl-3.0.16-h37729d1_0            |      |
| pip             | pkgs/main/win-64::pip-24.2-py38haa95532_0              |      |
| python          | pkgs/main/win-64::python-3.8.20-h8285438_0             |      |
| setuptools      | pkgs/main/win-64::setuptools-75.1.0-py38haa95532_0     |      |


```

```
Anaconda Prompt - conda in x + v
Proceed ([y]/n)? y

Downloading and Extracting Packages:
Preparing transaction: done
Verifying transaction: done
Executing transaction: done
#
# To activate this environment, use
#
#     $ conda activate rasa_env
#
# To deactivate an active environment, use
#
#     $ conda deactivate

(base) C:\Users\DELL>conda activate rasa_env
(rasa_env) C:\Users\DELL>conda install pip
Channels:
- defaults
Platform: win-64
Collecting package metadata (repodata.json): done
Solving environment: done

# All requested packages already installed.

(rasa_env) C:\Users\DELL>pip install rasa
Collecting rasa
  Downloading rasa-3.6.21-py3-none-any.whl.metadata (28 kB)
Collecting CacheControl<0.13.0,>=0.12.9 (from rasa)
  Downloading CacheControl-0.12.14-py2.py3-none-any.whl.metadata (2.2 kB)
Collecting PyJWT<3.0.0,>=2.0.0 (from PyJWT[crypto]<3.0.0,>=2.0.0->rasa)
  Downloading PyJWT-2.9.0-py3-none-any.whl.metadata (3.0 kB)
Collecting SQLAlchemy<1.5.0,>=1.4.0 (from rasa)
  Downloading SQLAlchemy-1.4.54-cp38-cp38-win_and64.whl.metadata (11 kB)
Collecting absl-py<1.5.0,>=0.9 (from rasa)
  Downloading absl-py-1.4.0-py3-none-any.whl.metadata (2.3 kB)
Collecting aio-pika<8.2.4,>=6.7.1 (from rasa)
  Downloading aio-pika-8.2.3-py3-none-any.whl.metadata (9.5 kB)
Collecting aiogram<2.26 (from rasa)
  Downloading aiogram-2.25.2-py3-none-any.whl.metadata (3.9 kB)
```

```
(base) C:\Users\DELL>pip install aiohttp==3.8.1
Collecting aiohttp==3.8.1
  Downloading aiohttp-3.8.1-cp38-cp38-win_and64.whl.metadata (7.3 kB)
Collecting attrs==17.3.0 (from aiohttp==3.8.1)
  Downloading attrs-25.1.0-py3-none-any.whl.metadata (10 kB)
Collecting charset-normalizer<3.0,>=2.0 (from aiohttp==3.8.1)
  Downloading charset-normalizer-2.1.1-py3-none-any.whl.metadata (11 kB)
Collecting multidict<7.0,>=4.0.5 (from aiohttp==3.8.1)
  Downloading multidict-6.1.0-cp38-cp38-win_and64.whl.metadata (5.1 kB)
Collecting async-timeout<5.0,>=4.0.0a3 (from aiohttp==3.8.1)
  Downloading async-timeout-4.0.3-py3-none-any.whl.metadata (4.2 kB)
Collecting yarl<2.0,>=1.0 (from aiohttp==3.8.1)
  Downloading yarl-1.15.2-cp38-cp38-win_and64.whl.metadata (58 kB)
Collecting frozenlist==1.1 (from aiohttp==3.8.1)
  Downloading frozenlist-1.5.0-cp38-cp38-win_and64.whl.metadata (14 kB)
Collecting aiosignal==1.1.2 (from aiohttp==3.8.1)
  Downloading aiosignal-1.3.1-py3-none-any.whl.metadata (4.0 kB)
Requirement already satisfied: typing-extensions==4.1.0 in c:\users\dell\anaconda3\envs\rasa_env\lib\site-packages (from multidict<7.0,>=4.0.5->aiohttp==3.8.1) (4.1.1)
Collecting idna==2.0 (from yarl<2.0,>=1.0->aiohttp==3.8.1)
  Using cached idna-2.10-py3-none-any.whl.metadata (9.4 kB)
Collecting propcache==0.2.0 (from yarl<2.0,>=1.0->aiohttp==3.8.1)
  Downloading propcache-0.2.0-cp38-cp38-win_and64.whl.metadata (7.9 kB)
Downloaded aiohttp-3.8.1-cp38-cp38-win_and64.whl (555 kB)
  356.9/555.0 kB 723.7 kB/s eta 0:00:00
Downloaded aiosignal-1.3.1-py3-none-any.whl (7.6 kB)
Downloaded async-timeout-4.0.3-py3-none-any.whl (5.7 kB)
Downloaded attrs-25.1.0-py3-none-any.whl (53 kB)
Downloaded charset-normalizer-2.1.1-py3-none-any.whl (39 kB)
Downloaded frozenlist-1.5.0-cp38-cp38-win_and64.whl (51 kB)
Downloaded multidict-6.1.0-cp38-cp38-win_and64.whl (28 kB)
Downloaded yarl-1.15.2-cp38-cp38-win_and64.whl (84 kB)
Using cached idna-3.10-py3-none-any.whl (70 kB)
Installing collected packages: propcache, multidict, yarl, aiosignal, aiohttp
Successfully installed aiohttp-3.8.1 aiosignal-1.3.1 async-timeout-4.0.3 attrs-25.1.0 charset-normalizer-2.1.1 frozenlist-1.5.0 idna-3.10 multidict-6.1.0 propcache-0.2.0 yarl-1.15.2

(rasa_env) C:\Users\DELL> pip install rasa==3.1.7 --no-deps
Collecting rasa==3.1.7
  Using cached rasa-3.1.7-py3-none-any.whl.metadata (23 kB)
Downloaded rasa-3.1.7-py3-none-any.whl (814 kB)
```

```
(base) C:\Users\DELL> pip install rasa==3.1.7 --no-deps
Collecting rasa==3.1.7
  Using cached rasa-3.1.7-py3-none-any.whl.metadata (23 kB)
Downloaded rasa-3.1.7-py3-none-any.whl (814 kB)
814.9/814.9 kB 737.0 kB/s eta 0:00:00
Installing collected packages: rasa
Successfully installed rasa-3.1.7
```

Activity 2.2: Setting up Rasa

```
Anaconda Prompt - conda in x + v

(base) C:\Users\DELL>conda create --name rasa_env python=3.8
Channels:
- defaults
Platform: win-64
Collecting package metadata (repodata.json): done
Solving environment: done

## Package Plan ##

  environment location: C:\Users\DELL\anaconda3\envs\rasa_env
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The following NEW packages will be INSTALLED:



| package         | pkgs/main/win-64                                        | size |
|-----------------|---------------------------------------------------------|------|
| ca-certificates | pkgs/main/win-64::ca-certificates-2025.2.25-haa95532_0  |      |
| libffi          | pkgs/main/win-64::libffi-3.4.4-hd77b12b_1               |      |
| openssl         | pkgs/main/win-64::openssl-3.0.16-h37729d1_0             |      |
| pip             | pkgs/main/win-64::pip-24.2-py38haa95532_0               |      |
| python          | pkgs/main/win-64::python-3.8.20-h8285438_0              |      |
| setuptools      | pkgs/main/win-64::setuptools-75.1.0-py38haa95532_0      |      |
| sqlite          | pkgs/main/win-64::sqlite-3.45.3-h2bbff1b_0              |      |
| vc              | pkgs/main/win-64::vc-14.42-haa95532_5                   |      |
| vs2015_runtime  | pkgs/main/win-64::vs2015_runtime-14.42.34433-hbf6602d_5 |      |
| wheel           | pkgs/main/win-64::wheel-0.44.0-py38haa95532_0           |      |



Proceed ([y]/n)? y

Downloading and Extracting Packages:
Preparing transaction: done
Verifying transaction: done
Executing transaction: done
#
# To activate this environment, use
#
#     $ conda activate rasa_env
#
# To deactivate an active environment, use
#
#     $ conda deactivate
```

Milestone 3: Data Collection & Preparation

This phase focuses on gathering, organizing, and structuring the necessary data to train the chatbot effectively. In the context of building a domain-specific chatbot for data science queries, the “data” refers to user queries (inputs), bot responses (outputs), and the flow of conversation (interactions). This stage is crucial for enabling Natural Language Understanding (NLU) and effective dialogue management in the RASA framework.

Activity 3.1: Creating User Queries

Objective: To create a diverse set of user inputs (also called training examples or utterances) that represent various ways users might ask questions related to data science.

Description: User queries are written examples of how people might interact with the chatbot. These utterances are used to train the chatbot’s intent classification model in RASA NLU. Each query is associated with a specific **intent**, such as asking for a definition, requesting a resource, or seeking tool-specific help.

```
! nlu.yml X
Users > dristi > Library > Containers > net.whatsapp.WhatsApp > Data > tmp > documents > D59AE003-3
1 version: "3.1"
2 nlu:
3
4 - intent: greet
5   examples: |
6     - hi
7     - hello
8     - hey there
9     - hey!
10    - good morning
11    - good afternoon
12    - what's up?
13    - yo
14    - hi there
15    - hey!
16
17 - intent: ask_course_info
18   examples: |
19     - What is the data science course about?
20     - Tell me about your Data Science & ML program
21     - Can you explain the data science course syllabus?
22     - I'd like course details
23     - What topics are covered in the course?
24     - What does the data science program include?
25     - What will I learn in this course?
26     - Can I know the course curriculum?
27     - What's covered in the DS course?
28     - What all will I learn if I enroll?
29     - Give me a brief of the syllabus
30
31 - intent: ask_duration
32   examples: |
33     - How long is the (data science)(course) bootcamp?
34     - How long is the (DS)(course) program?
35     - When does the (AI bootcamp)(course) start?
36     - How long is the (data science and ML)(course) course?
37     - Duration of data science & AI course?
38     - How many weeks does the DS course run?
39     - What's the length of the DS program?
40     - How long does the course last?
41     - What is the total duration?
42     - How many months is the course?
43
44 - intent: ask_eligibility
45   examples: |
46     - How much does the (data science)(course) course cost?
47     - What's the total fee for data science?
48     - Tell me the price of your DS bootcamp
49     - What is the total fee?
50     - How much do I have to pay?
51     - What are the charges for this course?
52     - Fee structure?
53     - Price of course?
54     - Do you offer EMI or discounts?
55
56 - intent: ask_admission_process
57   examples: |
58     - Am I eligible for the data science course?
59     - Eligibility criteria?
60     - Which background is required to join DS?
61     - Who can apply?
62     - What qualification is required?
63     - Do I need a tech background?
64     - Can non-CS students join?
65     - Is it open to everyone?
66     - I'm from commerce, can I do this?
67
68 - intent: ask_placement
69   examples: |
70     - How many months is the course?
71     - Course duration?
72     - Total time required for data science course?
73
74 - intent: ask_fees
75   examples: |
76     - How much does the (data science)(course) course cost?
77     - What's the total fee for data science?
78     - Tell me the price of your DS bootcamp
79     - What is the total fee?
80     - How much do I have to pay?
81     - What are the charges for this course?
82     - Fee structure?
83     - Price of course?
84     - Do you offer EMI or discounts?
85
86 - intent: ask_contact
87   examples: |
88     - Can I talk to a counsellor?
89     - What's the contact number?
90     - Provide email address
91     - Can someone call me?
92     - I want to speak to support
93     - Help me contact your team
94
95 - intent: goodbye
96   examples: |
97     - goodbye
98     - bye
99     - speak later
100    - see ya
101    - catch you later
102    - talk soon
103    - I'm done for now
104    - end chat
105
106 - intent: thankyou
107   examples: |
108     - Thanks!
109     - Thank you so much
110     - Cheers
111     - Appreciate it
112     - Much obliged
113     - Great, thanks!
114     - That helped!
115     - Nice, thank you!
116
117 - intent: ask_start_date
118   examples: |
119     - When does the next batch of data science start?
120     - What is the start date for the January 2026 batch?
121     - Tell me the upcoming start dates.
122     - I want to join next month.
123     - Dates for the new batch?
124     - Next batch schedule?
125     - When can I start?
126
127 - intent: ask_mode
128   examples: |
129     - Do you offer an (online)(mode) data science course?
130     - Are classes in-person or blended?
131     - I prefer online mode is that possible?
132     - Is it online?
133     - Can I take it offline?
134     - Do you offer hybrid classes?
135     - Is the course available remotely?
136     - Is it possible to attend from home?
137     - Do you offer online classes?
138     - (online)(mode)
139     - I want to take the course (offline)(mode)
140     - Is hybrid available?
141     - I prefer (hybrid)(mode)
142
143 - intent: ask_scholarship
144   examples: |
145     - Are there any scholarships available?
146     - Do you offer early-bird discounts?
147
148 - intent: deny
149   examples: |
150     - no
151     - nope
152     - not now
153     - maybe later
154     - I don't want that
155     - not interested
156     - no thanks
157     - nah
158     - cancel that
159
160 - intent: ask_fees
161   examples: |
162     - What's the fee for the (beginner)(course_level) program?
163     - Tell me about (advanced)(course_level) course fee
164     - I want the fee for (intermediate)(course_level)
165
166 - intent: ask_scholarship
167   examples: |
168     - Are there any scholarships available?
169     - Do you offer early-bird discounts?
170
171 - intent: ask_scholarship
172   examples: |
173     - Are there any scholarships available?
174     - Do you offer early-bird discounts?
175     - I'm looking for scholarship options.
176     - Can I get a fee waiver?
177     - Is any discount available?
178     - Can students get any support?
179
180 - intent: ask_topics
181   examples: |
182     - What does the curriculum include?
183     - Tell me about ML and NLP topics.
184     - Do you teach deep learning?
185     - Is statistics covered?
186     - What technologies will I learn?
187     - Show course subjects
188
189 - intent: ask_budget
190   examples: |
191     - My budget is $10k what can I do?
192     - I can spend up to $50,000 any course?
193     - I have a tight budget do you have cheaper options?
194     - 40000
195     - 30000
196     - I have 35,000
197     - I can pay 30k max
198     - Only have 25k budget
199     - My budget is $10k
200     - I can pay 30000
201     - 25000
202     - 40k
203     - I can afford up to $40,000
204
205 - intent: ask_practice
206   examples: |
207     - Does the course include hands-on coding?
208     - Will I get real-world projects?
209     - I want practical assignments.
210     - Are there mini projects?
211     - Will I get to build something?
212     - Is it theory or practical based?
213
214 - intent: affirm
215   examples: |
216     - yes
217     - yeah
218     - sure
219     - absolutely
220     - definitely
221     - of course
222     - please do
223     - okay
224     - alright
225     - yes
226     - I do
227     - I would
228     - I want to
229     - sounds good
230     - go ahead
231     - okay show me
232     - yes please
233     - why not
234     - obviously
235     - I guess so
236     - yup
237     - that's right
238     - correct
239     - cool
240     - totally
241
242 - intent: deny
243   examples: |
244     - no
245     - nope
246     - not now
247     - maybe later
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1129   examples: |
1130     - What's the fee for the (beginner)(course_level) program?
1131     - Tell me about (advanced)(course_level) course fee
1132     - I want the fee for (intermediate)(course_level)
1133
1134 - intent: ask_scholarship
1135   examples: |
1136     - Are there any scholarships available?
1137     - Do you offer early-bird discounts?
1138
1139 - intent: deny
1140   examples: |
1141     - no
1142     - nope
1143     - not now
1144     - maybe later
1145     - I don't want that
1146     - not interested
1147     - no thanks
1148     - nah
1149     - cancel that
1150
1151 - intent: ask_fees
1152   examples: |
1153     - What's the fee for the (beginner)(course_level) program?
1154     - Tell me about (advanced)(course_level) course fee
1155     - I want the fee for (intermediate)(course_level)
1156
1157 - intent: ask_scholarship
1158   examples: |
1159     - Are there any scholarships available?
1160     - Do you offer early-bird discounts?
1161
1162 - intent: deny
1163   examples: |
1164     - no
1165     - nope
1166     - not now
1167     - maybe later
1168     - I don't want that
1169     - not interested
1170     - no thanks
1171     - nah
1172     - cancel that
1173
1174 - intent: ask_fees
1175   examples: |
1176     - What's the fee for the (beginner)(course_level) program?
1177     - Tell me about (advanced)(course_level) course fee
1178     - I want the fee for (intermediate)(course_level)
1179
1180 - intent: ask_scholarship
1181   examples: |
1182     - Are there any scholarships available?
1183     - Do you offer early-bird discounts?
1184
1185 - intent: deny
1186   examples: |
1187     - no
1188     - nope
1189     - not now
1190     - maybe later
1191     - I don't want that
1192     - not interested
1193     - no thanks
1194     - nah
1195     - cancel that
1196
1197 - intent: ask_fees
1198   examples: |
1199     - What's the fee for the (beginner)(course_level) program?
1200     - Tell me about (advanced)(course_level) course fee
1201     - I want the fee for (intermediate)(course_level)
1202
1203 - intent: ask_scholarship
1204   examples: |
1205     - Are there any scholarships available?
1206     - Do you offer early-bird discounts?
1207
1208 - intent: deny
1209   examples: |
1210     - no
1211     - nope
1212     - not now
1213     - maybe later
1214     - I don't want that
12
```


Activity 3.2: Creating Bot Responses

Objective: To develop appropriate, accurate, and engaging responses for each user intent.

Description: Bot responses are stored under response templates in RASA and are triggered based on the intent identified by the NLU component. Responses should be informative, concise, and context-aware to ensure user satisfaction. Variations can be added for the same response to make conversations more natural.

```
! nlu.yml / domain.yml X
Users > distib > Library > Containers > net.whatsapp.WhatsApp > Data > tmp > documents > 31982AC3-0
1 version: "3.1"
2
3 intents:
4   - affirm
5   - greet
6   - ask_course_info
7   - ask_duration
8   - ask_fees
9   - ask_eligibility
10  - ask_admission_process
11  - ask_placement
12  - ask_prerequisite
13  - ask_contact
14  - goodbye
15  - thankyou
16  - ask_start_date
17  - ask_mode
18  - ask_scholarship
19  - ask_topics
20  - ask_budget
21  - ask_practice
22  - deny
23
24 entities:
25   - course
26   - duration
27   - location
28   - course_level
29   - mode
30   - start_date
31   - discount
32   - problem_topic
33   - budget
34
35 slots:
36   course:
37     type: text
38     mappings:
39       - type: from_entity
40       entity: course
41   duration:
42     type: text
43     mappings:
44       - type: from_entity
45       entity: duration
46
47 level:
48   type: text
49   mappings:
50     - type: from_entity
51     entity: course_level
52
53 budget:
54   type: text
55   influence_conversation: true
56   mappings:
57     - type: from_entity
58     entity: budget
59
60 responses:
61   utter_greet:
62     - text: "Hello 👋 I'm your Data Science course assistant. How can I help you today?"
63
64   utter_course_info:
65     - text: |
66       📖Data Science Program Overview📖
67       Our course includes:
68       - Python Programming
69       - Machine Learning (ML)
70       - Deep Learning (DL)
71       - Natural Language Processing (NLP)
72       - Data Visualization
73       - Real-world Projects & Case Studies
74
75       Want to know what topics we cover in detail?
76
77   utter_fees:
78     - text: |
79       📄Course Fee for {course}📄
80
81   utter_eligibility:
82     - text: |
83       📄The standard fee is ₹50,000📄
84       📄Scholarships & early-bird offers available!
85
86       Would you like me to check if you're eligible for any discount?
87
88   utter_admission_process:
89     - text: |
90       📋Admission Steps📋
91       1. Fill out the online application form
92       2. Attend a quick interview or screening
93       3. Make the payment to reserve your seat
94
95       Would you like the direct link to apply?
96
97   utter_prerequisite:
98     - text: |
99       📋Recommended Prerequisites📋
100      - Familiarity with Python
101      - Basic programming knowledge (preferably Python)
102      - Comfortable with high-school level math
103
104      Not sure if you're a good fit? I can help assess that too!
105
106   utter_placement:
107     - text: |
108       📋Placement Assistance📋
109       Yes! We offer:
110       - Resume building
111       - Mock interviews
112       - Industry connections
113       - Job referrals to hiring partners
114
115       Would you like recent placement stats?
116
117   utter_contact:
118     - text: |
119       📞Contact Support📞
120       Phone: +91-12345-67890
121       Email: support@example.com
122
123       Feel free to reach out anytime!
124
125   utter_goodbye:
126     - text: |
127       👋Goodbye! Come back anytime if you have more questions."
128
129   utter_thankyou:
130     - text: "You're welcome 🙏 Let me know if you need anything else."
131
132   utter_start_date:
133     - text: |
134       📅Next Batch Alerts📅
135       The upcoming batch starts on {(start_date)}.
136
137       Want to see the full calendar with weekend and weekday options?
138
139   utter_mode:
140     - text: |
141       📺We offer:
142       - Online (live + recorded)
143       - Offline (in select cities)
144       - Hybrid mode (flexible)
145
146       Which one works best for you?
147
148   utter_scholarship:
149     - text: |
150       🎓Scholarships & Offers🎓
151       - Early-bird discount: up to 15%
152       - Merit-based scholarship
153       - Group discounts available too!
154
155       Want to check if you qualify?
156
157   utter_topics:
158     - text: |
159       📚Topics You'll Master📚
160       - Python, Pandas, NumPy
161       - Machine Learning (supervised & unsupervised)
162       - Deep Learning (CNNs, RNNs)
163       - Natural Language Processing (NLP)
164       - Data Visualization (Tableau, PowerBI)
165       - Model deployment using Flask/Streamlit
166
167       Want the detailed syllabus as a PDF?
168
169   utter_budget:
170     - text: |
171       📊Based on your budget of {(budget)}, I recommend our 📄Certificate Program📄.
172       It includes:
173       - 8-week hands-on training
174       - Key ML modules
175       - 2 real-world projects
176
177       Shall I share the details?
178
179   utter_practice:
180     - text: |
181       📖Hands-on Learning📖
182       - 10+ coding assignments
183       - 3 major capstone projects
184
185       Would you like to see a sample project?
186
187   utter_ask_show_batches:
188     - text: "Would you like to see upcoming batch dates?"
189     buttons:
190       - title: "Yes"
191         payload: "/affirm"
192       - title: "No"
193         payload: "/deny"
194
195   actions:
196     - action_list_upcoming_batches
197     - action_course_duration
198     - action_provide_course_fee
199     - action_show_course_topics
200     - action_scholarship_info
201     - action_course_mode
202     - action_course_practice
203     - action_course_recommendation_by_budget
204     - action_provide_admission_link
205
206   session_config:
207     session_expiration_time: 60 # in minutes
208     carry_over_slots_to_new_session: true
209
210   114 utter_placement:
211     - text: |
212       📋Placement Assistance📋
213       Yes! We offer:
214       - Resume building
215       - Mock interviews
216       - Industry connections
217       - Job referrals to hiring partners
218
219       Would you like recent placement stats?
220
221   115 utter_contact:
222     - text: |
223       📞Contact Support📞
224       Phone: +91-12345-67890
225       Email: support@example.com
226
227       Feel free to reach out anytime!
228
229   116 utter_goodbye:
230     - text: |
231       👋Goodbye! Come back anytime if you have more questions."
232
233   117 utter_thankyou:
234     - text: "You're welcome 🙏 Let me know if you need anything else."
235
236   118 utter_start_date:
237     - text: |
238       📅Next Batch Alerts📅
239       The upcoming batch starts on {(start_date)}.
240
241       Want to see the full calendar with weekend and weekday options?
242
243   119 utter_mode:
244     - text: |
245       📺We offer:
246       - Online (live + recorded)
247       - Offline (in select cities)
248       - Hybrid mode (flexible)
249
250       Which one works best for you?
251
252   120 utter_scholarship:
253     - text: |
254       🎓Scholarships & Offers🎓
255       - Early-bird discount: up to 15%
256       - Merit-based scholarship
257       - Group discounts available too!
258
259       Want to check if you qualify?
260
261   121 utter_topics:
262     - text: |
263       📚Topics You'll Master📚
264       - Python, Pandas, NumPy
265       - Machine Learning (supervised & unsupervised)
266       - Deep Learning (CNNs, RNNs)
267       - Natural Language Processing (NLP)
268       - Data Visualization (Tableau, PowerBI)
269       - Model deployment using Flask/Streamlit
270
271       Want the detailed syllabus as a PDF?
272
273   122 utter_budget:
274     - text: |
275       📊Based on your budget of {(budget)}, I recommend our 📄Certificate Program📄.
276       It includes:
277       - 8-week hands-on training
278       - Key ML modules
279       - 2 real-world projects
280
281       Shall I share the details?
282
283   123 utter_practice:
284     - text: |
285       📖Hands-on Learning📖
286       - 10+ coding assignments
287       - 3 major capstone projects
288
289       Would you like to see a sample project?
```

Activity 3.3: Defining the Interaction Between the user and Chatbot

Objective: To design the conversational flow and dialogue structure that governs how the chatbot interacts with users across multiple turns.

Description: This activity involves defining the stories and rules that guide how the chatbot should respond based on user input and previous context. These are modelled using RASA Core.


```

! nlu.yml ! domain.yml ! stories.yml x ! rules.yml x
Users > dristi > Library > Containers > net.whatsapp.WhatsApp > Data > tmp > documents
1 version: "3.1"
2 stories:
3
4 - story: Course Inquiry Flow
5   steps:
6     - intent: greet
7     - action: utter_greet
8     - intent: ask_course_info
9     - action: utter_course_info
10    - intent: ask_topics
11    - action: action_show_course_topics
12    - intent: ask_duration
13    - action: action_course_duration
14    - intent: affirm
15    - action: action_list_upcoming_batches
16    - intent: ask_fees
17    - action: action_provide_course_fee
18    - intent: affirm
19    - action: action_scholarship_info
20    - intent: thankyou
21    - action: utter_thankyou
22    - intent: goodbye
23    - action: utter_goodbye
24
25 - story: Admission Help Flow
26   steps:
27     - intent: ask_admission_process
28     - action: action_provide_admission_link
29     - intent: ask_eligibility
30     - action: utter_eligibility
31     - intent: ask_prerequisite
32     - action: utter_prerequisite
33     - intent: ask_contact
34     - action: utter_contact
35     - intent: thankyou
36     - action: utter_thankyou
37     - intent: goodbye
38     - action: utter_goodbye
39
40 - story: Practice-Oriented Student
41   steps:
42     - intent: ask_practice

```

```

39
40 - story: Practice-Oriented Student
41   steps:
42     - intent: ask_practice
43     - action: action_course_practice
44     - intent: ask_topics
45     - action: action_show_course_topics
46     - intent: thankyou
47     - action: utter_thankyou
48
49 - story: Mode and Start Date Flow
50   steps:
51     - intent: ask_mode
52     - action: action_course_mode
53     - intent: ask_start_date
54     - action: action_list_upcoming_batches
55     - intent: affirm
56     - action: action_provide_admission_link
57     - intent: goodbye
58     - action: utter_goodbye
59
60 - story: Budget-Based Recommendation
61   steps:
62     - intent: ask_budget
63     - action: action_course_recommendation_by_budget
64     - intent: deny
65     - action: utter_goodbye
66
67 - story: Scholarship Interest After Fees
68   steps:
69     - intent: ask_fees
70     - action: action_provide_course_fee
71     - intent: affirm
72     - action: action_scholarship_info
73
74 - story: Placement Assistance Flow
75   steps:
76     - intent: ask_placement
77     - action: action_placement
78     - intent: ask_contact
79     - action: utter_contact

```

```

72   - action: action_scholarship_info
73
74 - story: Placement Assistance Flow
75   steps:
76     - intent: ask_placement
77     - action: utter_placement
78     - intent: ask_contact
79     - action: utter_contact
80     - intent: goodbye
81     - action: utter_goodbye
82
83 - story: User affirms after course duration
84   steps:
85     - intent: ask_duration
86     - action: action_course_duration
87     - intent: affirm
88     - action: action_list_upcoming_batches
89
90 - story: User affirms after fees
91   steps:
92     - intent: ask_fees
93     - action: action_provide_course_fee
94     - intent: affirm
95     - action: action_scholarship_info
96
97

```

```

! nlu.yml ! domain.yml ! stories.yml ! rules.yml x
Users > dristi > Library > Containers > net.whatsapp.WhatsApp > Data > tmp > documents
1 version: "3.1"
2 rules:
3
4 - rule: Greet
5   steps:
6     - intent: greet
7     - action: utter_greet
8
9 - rule: Say goodbye
10  steps:
11    - intent: goodbye
12    - action: utter_goodbye
13
14 - rule: Thank you
15  steps:
16    - intent: thankyou
17    - action: utter_thankyou
18
19 - rule: Provide course information
20  steps:
21    - intent: ask_course_info
22    - action: utter_course_info
23
24 - rule: Provide course duration
25  steps:
26    - intent: ask_duration
27    - action: action_course_duration
28
29 - rule: Provide fees
30  steps:
31    - intent: ask_fees
32    - action: action_provide_course_fee
33
34 - rule: Provide eligibility criteria
35  steps:
36    - intent: ask_eligibility
37    - action: utter_eligibility
38
39 - rule: Explain admission process
40  steps:
41    - intent: ask_admission_process
42    - action: action_provide_admission_link

```

```

38
39 - rule: Explain admission process
40  steps:
41    - intent: ask_admission_process
42    - action: action_provide_admission_link
43
44 - rule: List prerequisites
45  steps:
46    - intent: ask_prerequisite
47    - action: utter_prerequisite
48
49 - rule: Talk about placement support
50  steps:
51    - intent: ask_placement
52    - action: utter_placement
53
54 - rule: Share contact info
55  steps:
56    - intent: ask_contact
57    - action: utter_contact
58
59 - rule: Share upcoming batch dates
60  steps:
61    - intent: ask_start_date
62    - action: action_list_upcoming_batches
63
64 - rule: Mode of delivery
65  steps:
66    - intent: ask_mode
67    - action: action_course_mode
68
69 - rule: Share scholarship info
70  steps:
71    - intent: ask_scholarship
72    - action: action_scholarship_info
73
74 - rule: Share course topics
75  steps:
76    - intent: ask_topics
77    - action: action_show_course_topics
78
79 - rule: Share budget-based recommendation

```

```

78
79 - rule: Share budget-based recommendation
80  steps:
81    - intent: ask_budget
82    - action: action_course_recommendation_by_budget
83
84 - rule: Share practice/project details
85  steps:
86    - intent: ask_practice
87    - action: action_course_practice
88
89 # These are important multi-intent rules now properly handled
90
91 - rule: After budget suggestion, if user denies, exit
92  steps:
93    - intent: deny
94    - action: utter_goodbye
95
96 - rule: After course duration, if user denies, exit
97  steps:
98    - intent: deny
99    - action: utter_goodbye
100

```

Milestone 4: Training the Model

The training of the Rasa NLU (Natural Language Understanding) model represents a critical step in the creation of a chatbot system. This phase involves utilizing labelled training data to develop a model that can accurately determine user intents and extract meaningful entities from input text. The resulting model is essential for enabling the chatbot to interpret user inputs, generate appropriate responses, and maintain coherent dialogue flow.

Activity 4.1: Model Training

Objective: To initiate the training process for both the NLU and Core components of the chatbot using the `rasa train` command, enabling the creation of a fully integrated and functional conversational AI model.

Description: The `rasa train` command streamlines the training workflow by simultaneously training the NLU (Natural Language Understanding) and Core modules of the chatbot. This command helps build a cohesive model that can understand user inputs and manage conversations effectively, forming the backbone of a robust chatbot system.

1. Activate env:

```
● PS C:\Users\VIDIT KUMAR SINGH\Rasa_Project> rasa_env\Scripts\activate
○ (rasa_env) PS C:\Users\VIDIT KUMAR SINGH\Rasa_Project> █
```

2. Train Model:

```
● (rasa_env) PS C:\Users\VIDIT KUMAR SINGH\PROJECT AI> rasa train
C:\Users\VIDIT KUMAR SINGH\PROJECT AI\rasa_env\lib\site-packages\rasa\core\tracker_store.py:1048: MovedIn20Warn
ing: Deprecated API features detected! These feature(s) are not compatible with SQLAlchemy 2.0. To prevent inco
mpatible upgrades prior to updating applications, ensure requirements files are pinned to "sqlalchemy<2.0". Set
environment variable SQLALCHEMY_WARN_20=1 to show all deprecation warnings. Set environment variable SQLACHE
MY_SILENCE_UBER_WARNING=1 to silence this message. (Background on SQLAlchemy 2.0 at: https://sqlalche.me/e/b8d9
)
Base: DeclarativeMeta = declarative_base()
C:\Users\VIDIT KUMAR SINGH\PROJECT AI\rasa_env\lib\site-packages\rasa\shared\utils\validation.py:131: Deprecati
onWarning: pkg_resources is deprecated as an API. See https://setuptools.pypa.io/en/latest/pkg_resources.html
import pkg_resources
C:\Users\VIDIT KUMAR SINGH\PROJECT AI\rasa_env\lib\site-packages\pkg_resources\__init__.py:3154: DeprecationWar
ning: Deprecated call to `pkg_resources.declare_namespace('mpl_toolkits')`.
Implementing implicit namespace packages (as specified in PEP 420) is preferred to `pkg_resources.declare_names
pace`. See https://setuptools.pypa.io/en/latest/references/keywords.html#keyword-namespace-packages
declare_namespace(pkg)
C:\Users\VIDIT KUMAR SINGH\PROJECT AI\rasa_env\lib\site-packages\pkg_resources\__init__.py:3154: DeprecationWar
ning: Deprecated call to `pkg_resources.declare_namespace('google')`.
Implementing implicit namespace packages (as specified in PEP 420) is preferred to `pkg_resources.declare_names
pace`. See https://setuptools.pypa.io/en/latest/references/keywords.html#keyword-namespace-packages
declare_namespace(pkg)
C:\Users\VIDIT KUMAR SINGH\PROJECT AI\rasa_env\lib\site-packages\tensorflow\python\framework\dtypes.py:246: Dep
recationWarning: `np.bool8` is a deprecated alias for `np.bool_`. (Deprecated NumPy 1.24)
np.bool8: (False, True),
C:\Users\VIDIT KUMAR SINGH\PROJECT AI\rasa_env\lib\site-packages\rasa\engine\recipes\recipe.py:35: FutureWarnin
g: From Rasa Open Source 4.0.0 onwards it will be required to specify a recipe in your model configuration. Def
aulting to recipe 'default.v1'.
rasa.shared.utils.io.raise_deprecation_warning(
```

```

C:\Users\VIDIT KUMAR SINGH\PROJECT AI\rasa_env\lib\site-packages\rasa\engine\recipes\recipe.py:35: FutureWarning: From Rasa Open Source 4.0.0 onwards it will be required to specify a recipe in your model configuration. Defaulting to recipe 'default.v1'.
  rasa.shared.utils.io.raise_deprecation_warning(
C:\Users\VIDIT KUMAR SINGH\PROJECT AI\rasa_env\lib\site-packages\rasa\shared\utils\io.py:99: UserWarning: Entity entity 'problem_topic' has only 1 training examples! The minimum is 2, because of this the training may fail.
2025-07-04 23:47:38 INFO      rasa.engine.training.hooks - Restored component 'RegexFeaturizer' from cache.
2025-07-04 23:47:38 INFO      rasa.engine.training.hooks - Restored component 'LexicalSyntacticFeaturizer' from cache.
2025-07-04 23:47:38 INFO      rasa.engine.training.hooks - Restored component 'CountVectorsFeaturizer' from cache.
2025-07-04 23:47:38 INFO      rasa.engine.training.hooks - Restored component 'CountVectorsFeaturizer' from cache.
2025-07-04 23:47:38 INFO      rasa.engine.training.hooks - Starting to train component 'EntitySynonymMapper'.
2025-07-04 23:47:39 INFO      rasa.engine.training.hooks - Finished training component 'EntitySynonymMapper'.
2025-07-04 23:47:39 INFO      rasa.engine.training.hooks - Starting to train component 'ResponseSelector'.
C:\Users\VIDIT KUMAR SINGH\PROJECT AI\rasa_env\lib\site-packages\rasa\utils\train_utils.py:528: UserWarning: constrain_similarities is set to 'False'. It is recommended to set it to 'True' when using cross-entropy loss.
  rasa.shared.utils.io.raise_warning(
2025-07-04 23:47:39 INFO      rasa.nlu.selectors.response_selector - Retrieval intent parameter was left to its default value. This response selector will be trained on training examples combining all retrieval intents.
2025-07-04 23:47:39 INFO      rasa.engine.training.hooks - Finished training component 'ResponseSelector'.
2025-07-04 23:47:39 INFO      rasa.engine.training.hooks - Restored component 'DIETClassifier' from cache.
2025-07-04 23:47:39 INFO      rasa.engine.training.hooks - Restored component 'MemoizationPolicy' from cache.
2025-07-04 23:47:39 INFO      rasa.engine.training.hooks - Restored component 'RulePolicy' from cache.
2025-07-04 23:47:39 INFO      rasa.engine.training.hooks - Restored component 'TEDPolicy' from cache.
Your Rasa model is trained and saved at 'models\20250704-234736-average-pulse.tar.gz'.

```

Activity 4.2: Save the Model

Objective: To store the trained Rasa model in the designated project directory, making it available for deployment and integration with messaging platforms or production environments.

Description: Once the training process is successfully completed and the model performs as expected, Rasa automatically saves it in the project's models directory. These saved models are ready to be used for chatbot deployment, allowing seamless integration with various platforms and enabling real-time user interaction.

▼ models		
≡	20250704-202242-burning-amortization.tar.gz	U
≡	20250704-205228-great-biography.tar.gz	U
≡	20250704-210327-mean-chinon.tar.gz	U
≡	20250704-211250-frosty-grappa.tar.gz	U
≡	20250704-212706-tense-avocado.tar.gz	U
≡	20250704-214136-greedy-class.tar.gz	U
≡	20250704-214806-molto-grappa.tar.gz	U
≡	20250704-223005-ancient-napalm.tar.gz	U
≡	20250704-223748-charitable-drive.tar.gz	U
≡	20250704-224334-glowing-desert.tar.gz	U
≡	20250704-225241-cerulean-energy.tar.gz	U

Milestone 5: Testing the Model

Activity 5.1: Testing

Objective: To test and validate the performance of the trained chatbot model by interacting with it using the RASA shell command, ensuring accurate intent recognition and appropriate response generation.

Description: Model testing is a crucial phase in chatbot development that involves evaluating the chatbot's responses to various user inputs. Using the RASA shell command, developers can simulate real-time conversations with the chatbot through a command-line interface. This interactive testing environment helps verify the model's ability to understand inputs, follow dialogue flows, and respond appropriately. It also allows for quick identification of issues and facilitates fine-tuning of the model's behaviour before deployment.

```
(rasa_env) PS C:\Users\VIDIT KUMAR SINGH\PROJECT AI> rasa shell
C:\Users\VIDIT KUMAR SINGH\PROJECT AI\rasa_env\lib\site-packages\rasa\core\tracker_store.py:1048: MovedIn20Warning:
  Deprecated API features detected! These feature(s) are not compatible with SQLAlchemy 2.0. To prevent incompatible upgrades prior to updating applications, ensure requirements files are pinned to "sqlalchemy<2.0". Set environment variable SQLALCHEMY_WARN_20=1 to show all deprecation warnings. Set environment variable SQLALCHEMY_SILENCE_UBER_WARNING=1 to silence this message. (Background on SQLAlchemy 2.0 at: https://sqlalche.me/e/b8d9)
)
Base: DeclarativeMeta = declarative_base()
C:\Users\VIDIT KUMAR SINGH\PROJECT AI\rasa_env\lib\site-packages\rasa\shared\utils\validation.py:131: DeprecationWarning: pkg_resources is deprecated as an API. See https://setuptools.pypa.io/en/latest/pkg_resources.html
  import pkg_resources
C:\Users\VIDIT KUMAR SINGH\PROJECT AI\rasa_env\lib\site-packages\pkg_resources\__init__.py:3154: DeprecationWarning: Deprecated call to `pkg_resources.declare_namespace('mpl_toolkits')`. Implementing implicit namespace packages (as specified in PEP 420) is preferred to `pkg_resources.declare_namespace('')`. See https://setuptools.pypa.io/en/latest/references/keywords.html#keyword-namespace-packages
  declare_namespace(pkg)
C:\Users\VIDIT KUMAR SINGH\PROJECT AI\rasa_env\lib\site-packages\pkg_resources\__init__.py:3154: DeprecationWarning: Deprecated call to `pkg_resources.declare_namespace('google')`. Implementing implicit namespace packages (as specified in PEP 420) is preferred to `pkg_resources.declare_namespace('')`. See https://setuptools.pypa.io/en/latest/references/keywords.html#keyword-namespace-packages
  declare_namespace(pkg)
C:\Users\VIDIT KUMAR SINGH\PROJECT AI\rasa_env\lib\site-packages\tensorflow\python\framework\dtypes.py:246: DeprecationWarning: `np.bool8` is a deprecated alias for `np.bool_`. (Deprecated NumPy 1.24)
  np.bool8: (False, True),
C:\Users\VIDIT KUMAR SINGH\PROJECT AI\rasa_env\lib\site-packages\sanic_cors\extension.py:39: DeprecationWarning: distutils Version classes are deprecated. Use packaging.version instead.
```

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C:\Users\VIDIT KUMAR SINGH\PROJECT AI\rasa_env\lib\site-packages\tensorflow\python\framework\dtypes.py:246: DeprecationWarning: `np.bool8` is a deprecated alias for `np.bool_`. (Deprecated NumPy 1.24)
  np.bool8: (False, True),
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```

Activity 5.2: Output

While interacting with the chatbot through the RASA shell, you have the opportunity to experiment with a variety of user inputs, including unexpected or edge-case scenarios. This allows you to thoroughly assess how well the chatbot handles different types of conversations. In particular, you can evaluate the effectiveness of fallback mechanisms and error-handling strategies by deliberately entering ambiguous, irrelevant, or out-of-scope queries.

This hands-on approach is incredibly valuable for identifying gaps in intent recognition, detecting incorrect responses, and observing how the chatbot navigates complex dialogue paths. The interactive nature of the RASA shell provides immediate feedback, enabling you to debug and refine your model on the spot. Through repeated testing and iteration, you can improve the overall accuracy, reliability, and conversational flow of your chatbot, ensuring it delivers a more natural and user-friendly experience when deployed in real-world scenarios.

```
2025-07-04 23:50:30 INFO      root - Rasa server is up and running.
Bot loaded. Type a message and press enter (use '/stop' to exit):
Your input -> hello
Hello 🤖! I'm your Data Science course assistant. How can I help you today?
Your input -> Tell me about the course
▣ **Data Science Program Overview**
Our course includes:
• Python Programming
• Machine Learning (ML)
• Deep Learning (DL)
• Natural Language Processing (NLP)
• Data Visualization
• Real-world Projects & Case Studies
Your input -> What does the curriculum include
📖 Our curriculum includes:
🔥 Python Programming
📊 Statistics & Data Analysis
🧠 Machine Learning Algorithms
🧠 Deep Learning (CNNs, RNNs)
🗣️ Natural Language Processing
🏠 Real-World Capstone Projects
Your input -> What will be the duration?
🕒 The *Data Science* course runs for **12 weeks**.
Your input -> What will be the start date?
Here are our upcoming batch start dates:
- 📅 July 15, 2025
- 📅 September 1, 2025
- 📅 November 10, 2025
Your input -> How much the DS course cost?
Please specify whether you're interested in beginner, intermediate, or advanced level.
Your input -> advanced
```

```
Please specify whether you're interested in beginner, intermediate, or advanced level.
Your input -> advanced
The fee for our Advanced level course is ₹50,000. Scholarships and EMI options are also available.
Your input -> Any scholarships?
👉 **Scholarships & Offers**
- Early Bird: 10% off
- Merit Scholarship: Up to 25% for top candidates
- Group Discount: For 3+ enrollments
- EMI Plans also available for flexible payments.
Your input -> Thanks
You're welcome 😊 Let me know if you need anything else.
Your input ->
```